



TEAM 3

Traffic Vision

AI

Team Members

- | | |
|-----------------|-----------------------|
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UNDERSTAND THE PROJECT STATEMENT

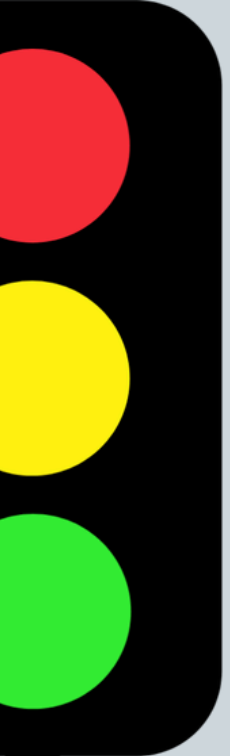
This is a web app where a city traffic department can log in, see live traffic conditions on a map/dashboard, get AI predictions about upcoming congestion, receive suggested alternate routes, and get alerted about accidents or jams — all backed by a database of traffic data and a simple ML model.



OBJECTIVES

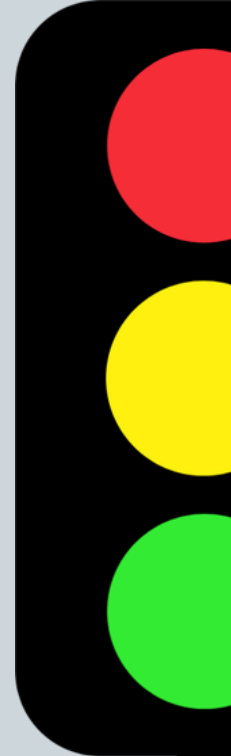
- Provide real-time visibility into traffic conditions across monitored roads
- Predict congestion before it happens using historical and live data (AI/ML)
- Recommend alternate routes to reduce travel time during congestion
- Alert traffic operators and the public about accidents, closures, and jams
- Give traffic authorities analytics (trends, heatmaps) to plan long-term improvements





USER ROLES

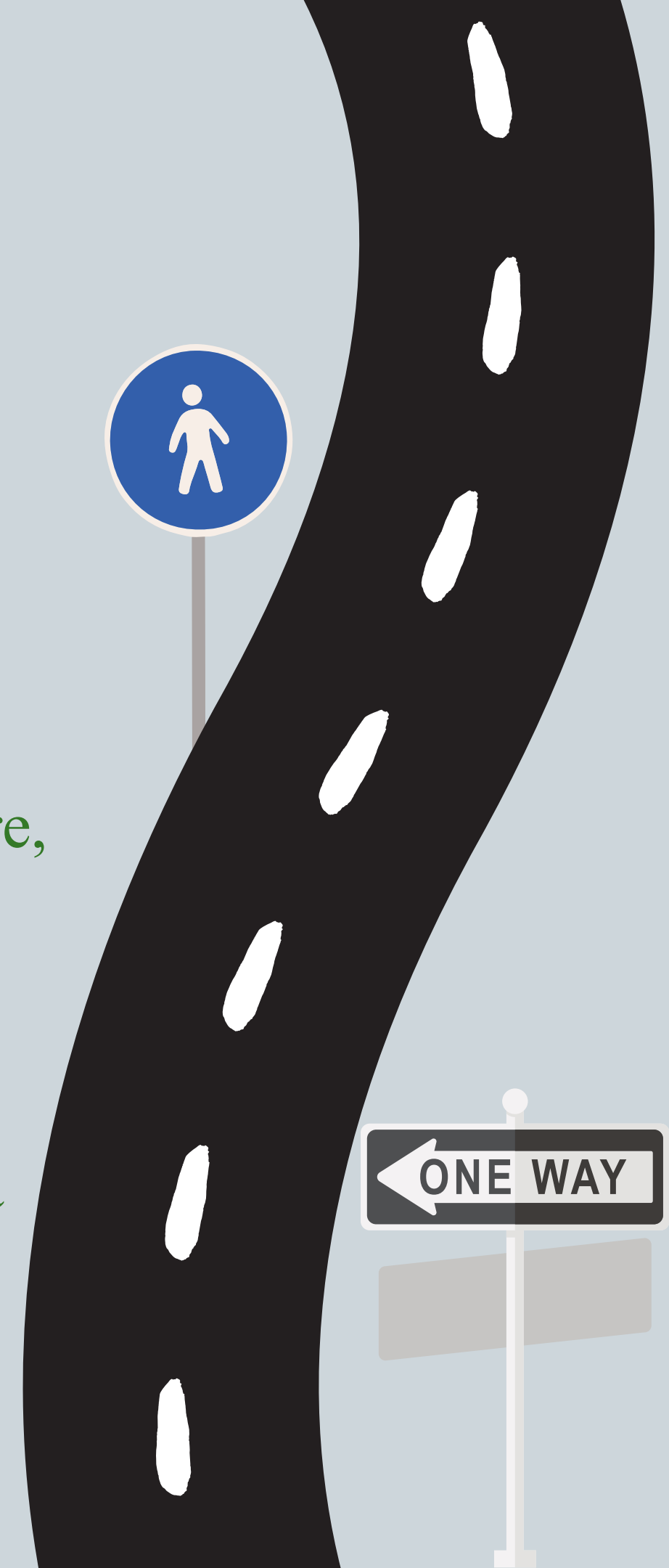
Role	Access	Key actions
Admin	Full access	Manage users, configure roads/sensors, view all analytics
Traffic operator	Operational access	Monitor live dashboard, respond to alerts, mark incidents
Public / commuter	Read-only, limited	View live map, get route suggestions, receive alerts



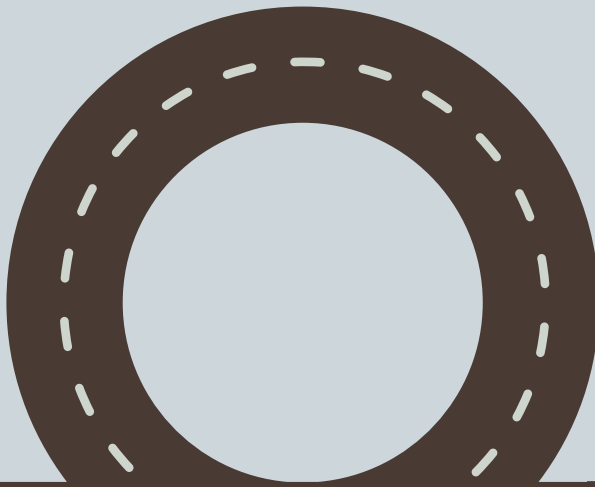
RESEARCH ON SMART TRAFFIC

DASHBOARDS

- Modern control-room dashboards combine a live map (color-coded green/yellow/orange/red for congestion levels) with a side panel showing incidents, vehicle counts, and infrastructure status such as bridges and underpasses.
- Mature intelligent traffic systems bring together the traffic management centre, command dashboards, alert workflows, and operator interfaces into one operating core, so the dashboard is not just passive monitoring but supports active decisions.
- Real intelligent traffic management platforms typically offer multimodal data integration, AI-based route planning, an efficiency dashboard, and urban analytics — a useful checklist to compare this project against.



RESEARCH ON CONGESTION MONITORING

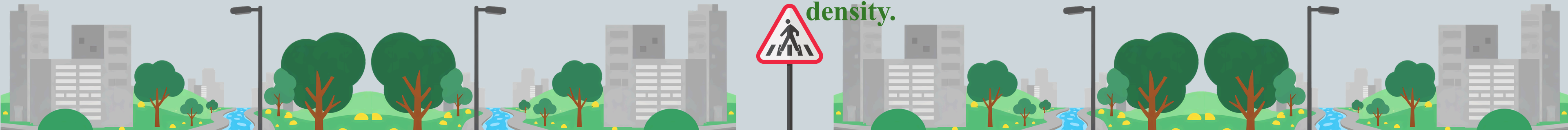


•GPS/probe-based — using GPS traces from vehicles or phones (as Google/Waze do); a road segment is often flagged congested when average speed drops significantly below normal (some studies use under 10 km/h as a threshold).

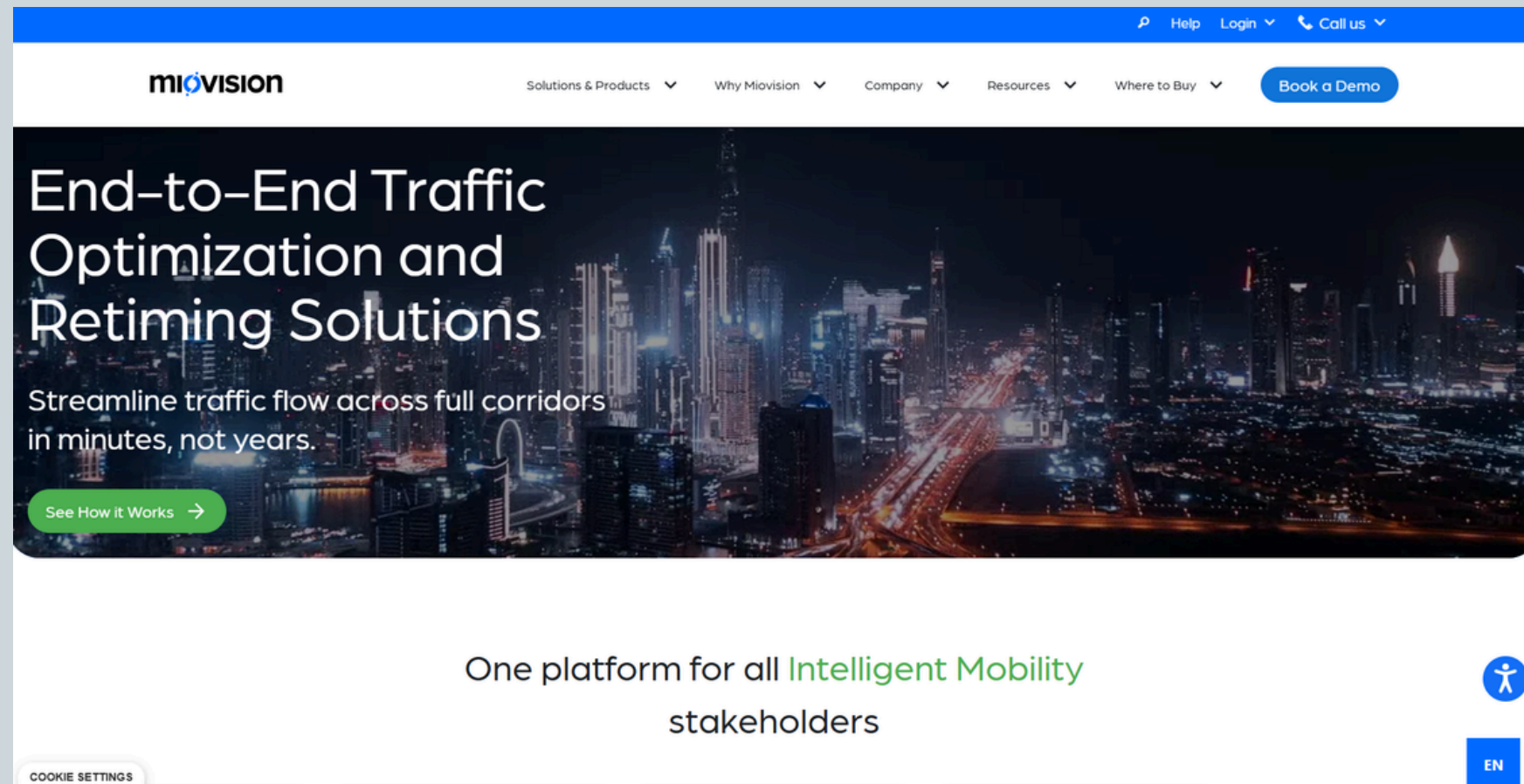
• Fixed roadside sensors — inductive loop sensors, CCTV/video analytics, infrared, radar, and RFID sensors installed on roads; good accuracy but limited geographic coverage.

•Computer vision / deep learning — models such as YOLO or Faster R-CNN detect and count vehicles from camera feeds, then classify congestion as light, medium, or heavy based on density.

•Sensor fusion — combining radar, GPS, and cameras together, since no single sensor type is fully reliable on its own in real-world conditions.

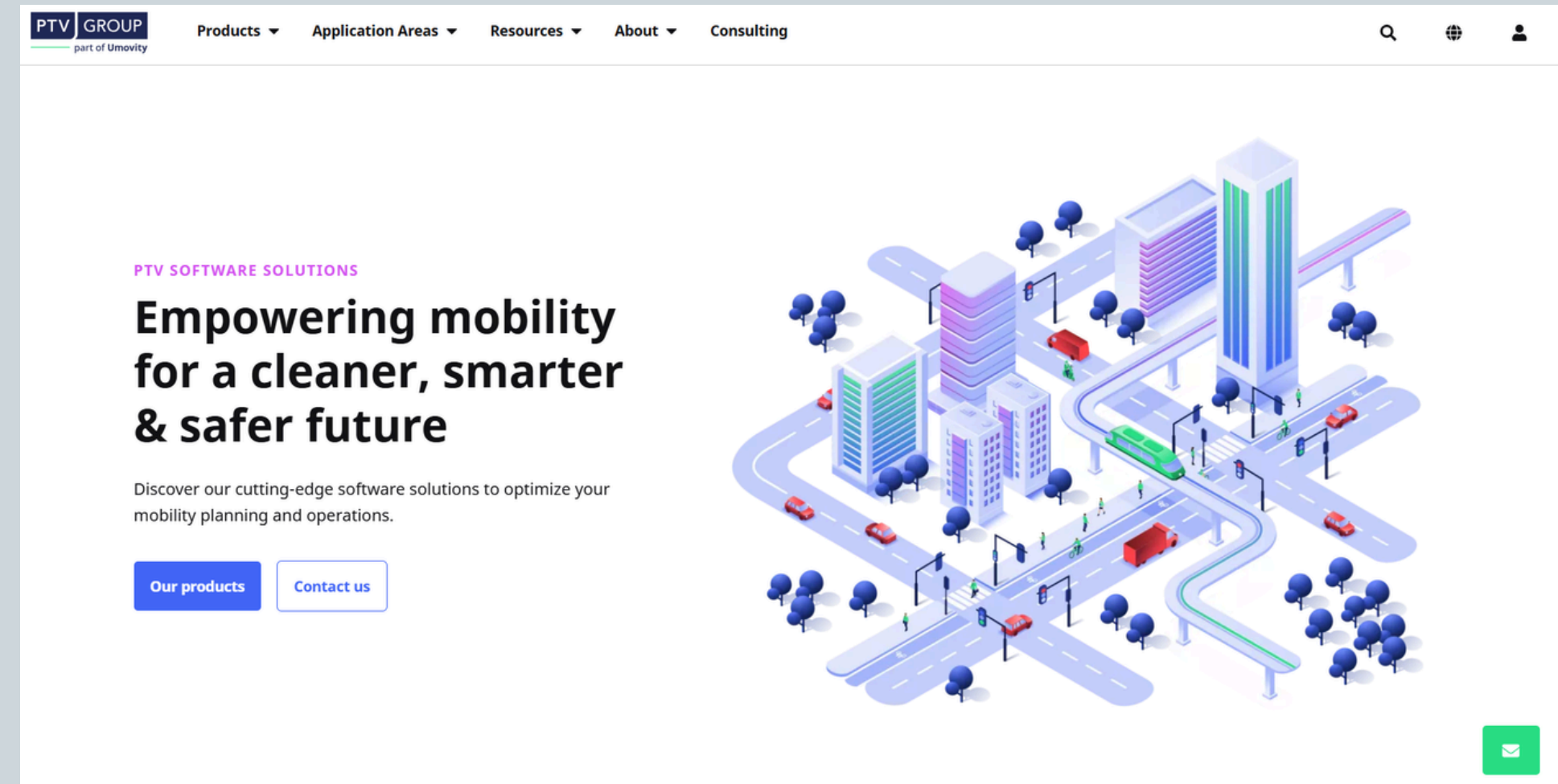


UI DESIGN INSPIRATION



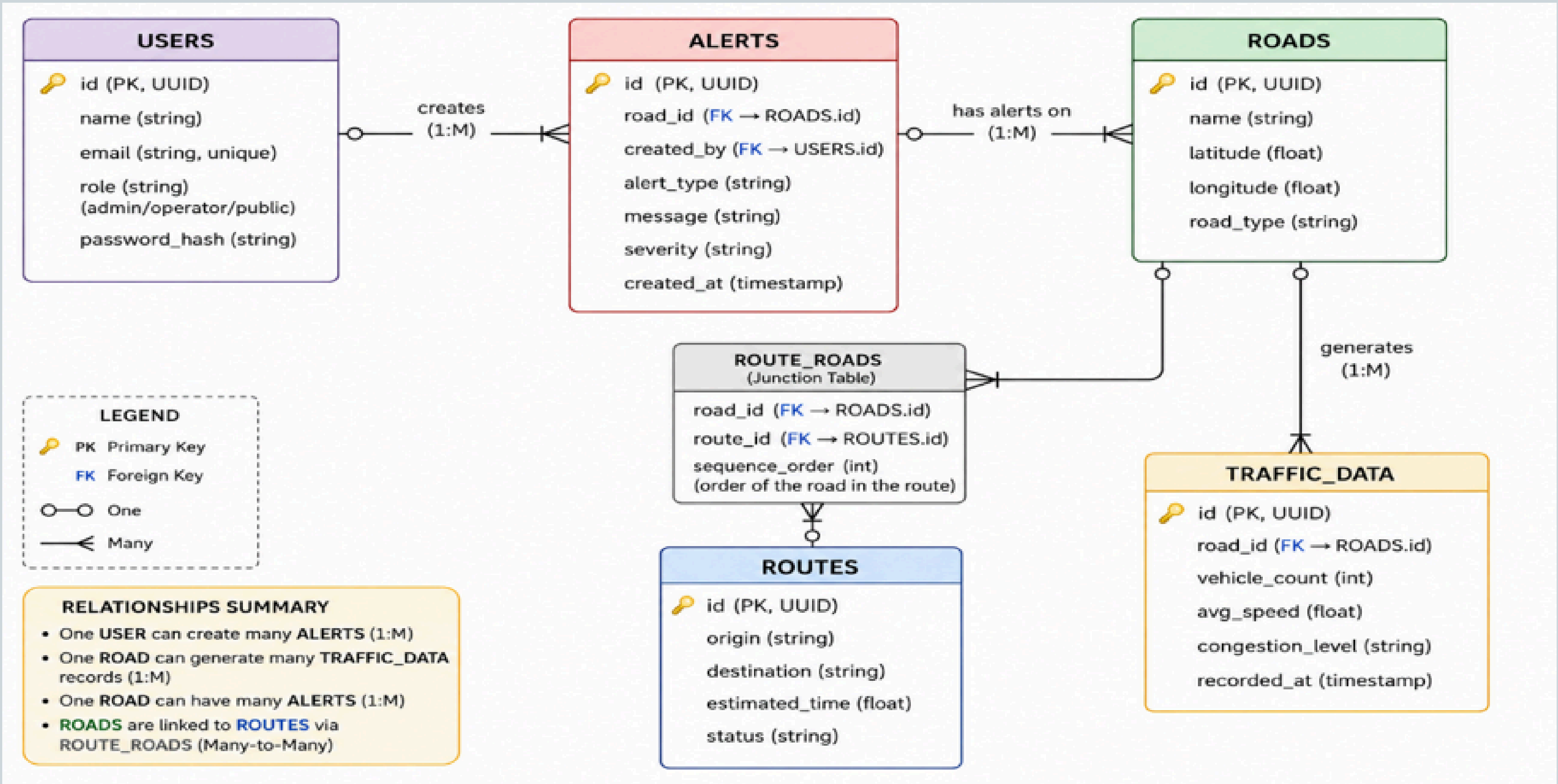
<https://miovision.com/>

Reference dashboards and platforms used to guide the design of the TrafficVision AI interface.

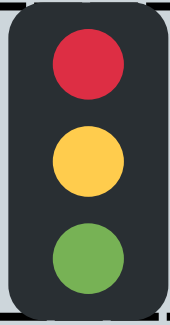


<https://www.ptvgroup.com/en>

ER DIAGRAM



TRAFFIC MANAGEMENT WORKFLOWS



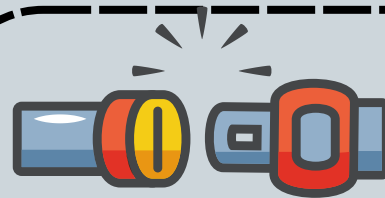
1. Live Monitoring: Sensor data → stored in database → shown on live dashboard.



Congestion Prediction: Historical + live data → ML model → congestion forecast displayed



1. Route Recommendation: Request received → routes ranked by congestion → best route returned.



1. Alerts & Incidents: Threshold crossed / incident reported → alert sent → operator responds.



CONCLUSION

1. Defined clear project objectives and smart traffic management workflows
2. Designed system architecture and database schema for scalability
3. Created UI wireframes and workflow plans for user-centric design
4. Set up frontend and backend development environments
5. Implemented secure authentication with role-based access control
6. Built a functional live traffic monitoring dashboard
7. Developed initial congestion tracking workflows
8. Gained hands-on understanding of system design, database architecture, and full-stack setup
9. Delivered a working authentication and traffic monitoring system, ready for advanced feature development in the next milestone



Safe

Always

THANK
YOU

